STAFF REVIEW · H446 185 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. [2,4,5].

2. 3.

3. LIFO: last in, first out.

4. Track whether any swap occurred; if none occurred during a complete pass, stop.

I Do — source worked steps

1. Predict first — Identify the shifts needed.

2. Model solution — Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].

We Do — guided CHECK

1. Model solution
9 moves one index right; 2 stays. Insert 5 at index 1 → [2,5,9,7].

You Do — source mark schemes

1. Shift 6, then insert 3 → [2,3,6,8].

2. After key 1 → [1,5,4]; after key 4 → [1,4,5].

3. Few values need shifting, so its best-case time is O(n), though worst case is $O(n^2)$.

Exit Ticket — source answer

1. False. It commonly shifts larger values and places the key in the gap.